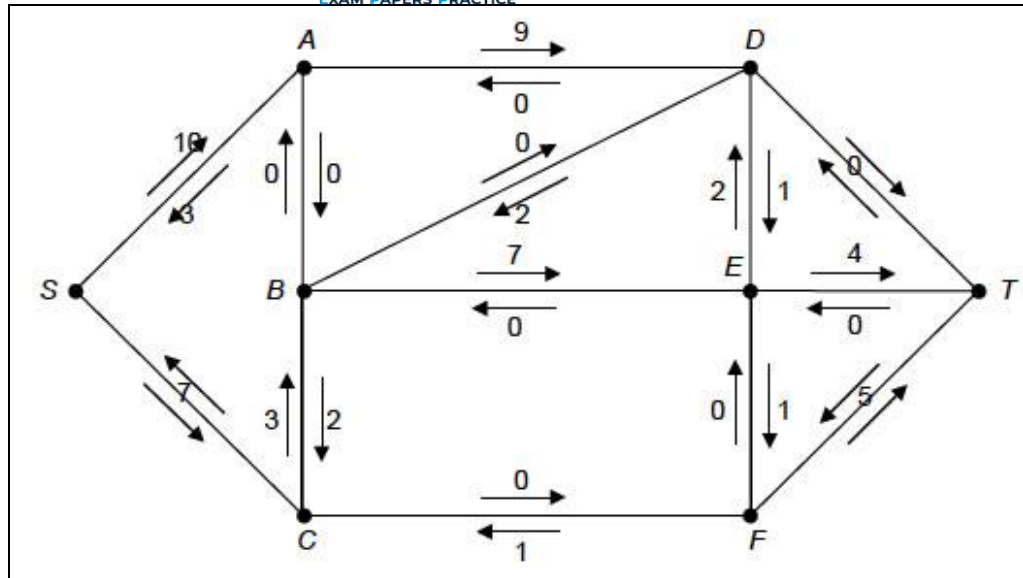


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Explains correctly using maximum flow into and minimum flow out of A	AO2.4	E1	Max flow into vertex $A = 7$ Min flow out of vertex $A = 4 + 3 = 7$ As flow into $A =$ flow out of A , both AD and AB must be at the lower capacity
(i)				
(ii)	Explains correctly using maximum flow into and minimum flow out of E	AO2.4	E1	$AD = 4$ from (a)(i), so $DE = 1$ and $DT = 3$. Since flow out of vertex E is at least 2, and $DE = 1$, BE must be at its upper capacity of 1.
(b)	Explains correctly the statement using a cut in the network	AO2.4	E1	Min flow across the cut $\{S, A, B, C\} \setminus \{D, E, G, T\}$ $= AD + BE + \min(BG) + \min(CG)$ $= 4 + 1 + 5 + 2$ $= 12 > 11$

(c)	Finds a potential increase and decrease for each arc with a value on the two arrows for each arc, with SA, SC, AB, BC and CF correct.	AO1.1a	M1	See diagram below
(i)	Determines correctly the potential increase and decrease for each arc with values on all arrows correct.	AO1.1b	A1	



(ii)	Finds correctly one augmenting path and flow	AO1.1b	B1	<table><tr><th>Augmenting Path</th><th>Flow</th></tr><tr><td>SADEFT</td><td>1</td></tr><tr><td>SCBET</td><td>3</td></tr><tr><td>SADBET</td><td>1</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	Augmenting Path	Flow	SADEFT	1	SCBET	3	SADBET	1				
	Augmenting Path	Flow														
	SADEFT	1														
SCBET	3															
SADBET	1															
Finds correctly two augmenting paths and the flows	AO1.1b	B1														
Finds correctly three (or four) augmenting paths and the flows, and clearly states the maximum flow in context with units	AO3.2a	B1	Maximum flow through the network is 22 litres per second													
(iii)	Constructs a rigorous mathematical proof using the value of a cut and the maximum flow-minimum cut theorem (to achieve this mark, the student must clearly show the cut they are calculating the value of and clearly state that the value of this cut is equal to the value of the flow in (c)(ii) and then conclude that the flow is maximal by the maximum flow-minimum cut theorem)	AO2.1	R1	$\{S, A, B, C, D, E\} / \{F, T\} = 7 + 3 + 9 + 3 = 22$ As the flow (22) is equal to the value of the cut (22), the maximum flow is 22 by the maximum flow-minimum cut theorem.												
(iv)	Argues that, as 7 litres per second are initially flowing through node E, only a further 2 litres per second can flow through node E	AO2.4	R1	Flow through network can only increase by 2 litres per second as DT and CF are already saturated Therefore restricted capacity node reduces the maximum flow to $17 + 2 = 19$ litres per second												

Interprets the impact of the restricted capacity node, concluding that the maximum flow is reduced to 19 litres per second	AO3.A	R1
Total 11 marks		

Q2.

(a) $ABEH$ 8

$ACFH$ 5

$ADGH$ 11

B1

1

(b) (i) $ACEH$ 2

One correct route and flow

M1

$ACGH$ 4

At least one other correct

A1

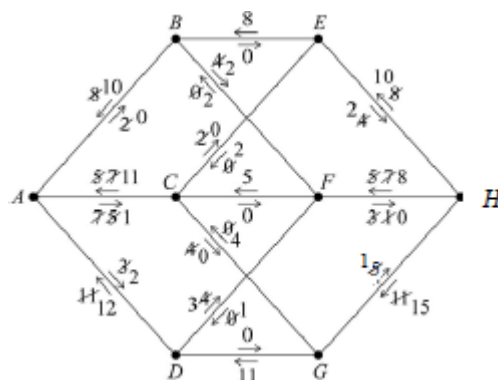
Either $ADFH$ 1 and $ABFH$ 2

All correct

A1

Or $ADFH$ 3

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Forward and back flows on diagram

M1

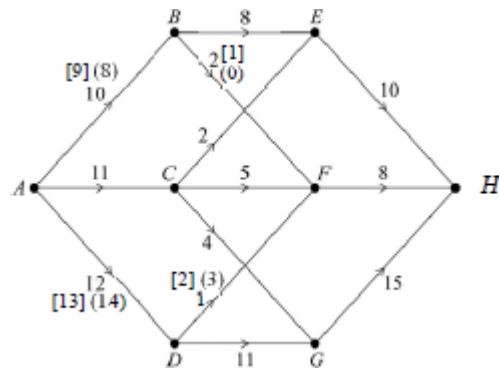
All correct

A1

5

(ii) Max flow 33

B1



OE

B1

2

(c) Cut through BE, CE, FH, CG, DG

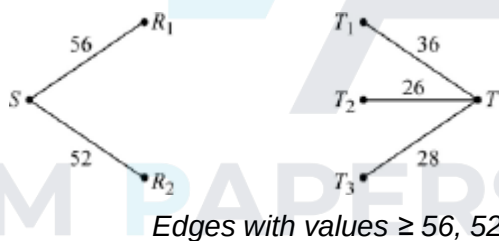
B1

1

[9]

Q3.

(a)



Edges with values $\geq 56, 52$

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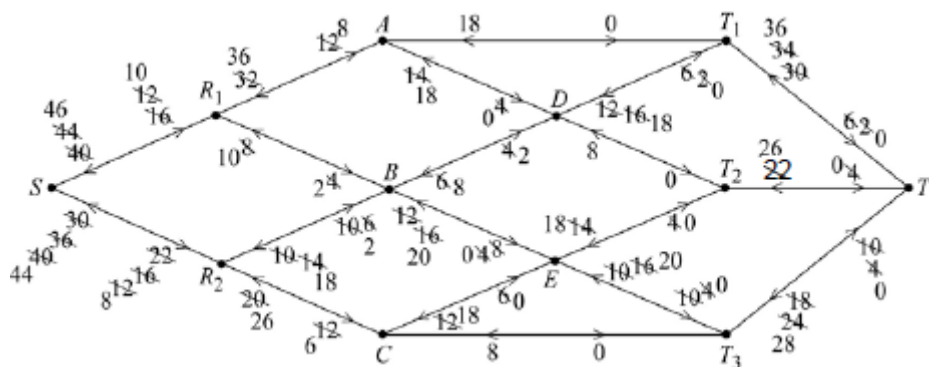
B1

Edges with values $\geq 36, 26, 28$

B1

2

(b)(i)



initial diagram with forward / back flows

M1

Fully correct diagram

A1

$S R_1 A D T_1 T \quad 4$

One correct path and flow

M1

$S R_1 B D T_1 T \quad 2$

At least one other correct path and flow

A1

$S R_2 C E T_3 T \quad 6$

$S R_2 B E T_2 T \quad 4$

all correct

A1

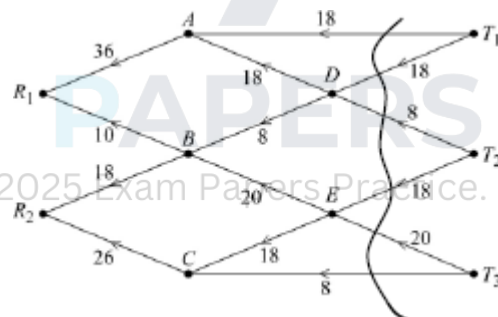
$S R_2 B E T_3 T \quad 4$

(ignore connections to S and T)

5

(ii) Max flow 90

B1



B1

2

(c) Cut through (shown)

PI by correct list

B1

$AT_1, DT_1, DT_2, ET_2, ET_3, CT_3$

B1

OE

2

[11]

Q4.

(a) $10 + 13 - 1 + 17$

3 values added and -1 (condone one slip)

M1

$= 39$

A1

2

(b) (i) $DE \quad 12$

on Figure 1

B1

$FG \quad 7$

B1

2

(ii)

arc	forward	backward
SA	3	1
AB	1	1
BT	0	1
SC	0	2
CA	0	1
AD	0	1
CD	1	1
DE	1	2
BE	1	3
ET	2	3
SF	1	1
FC	1	2
FD	1	0
FG	1	1
DG	0	1
EG	2	1
GT	1	3
	2	

at least 6 pairs correct on Figure 2 (must have arrows)

M1

all correct

A1

2

(iii) **Table**

Path	Extra Flow

SABET	1
SFDGT	1
SACDGT	1

1 correct path and extra flow

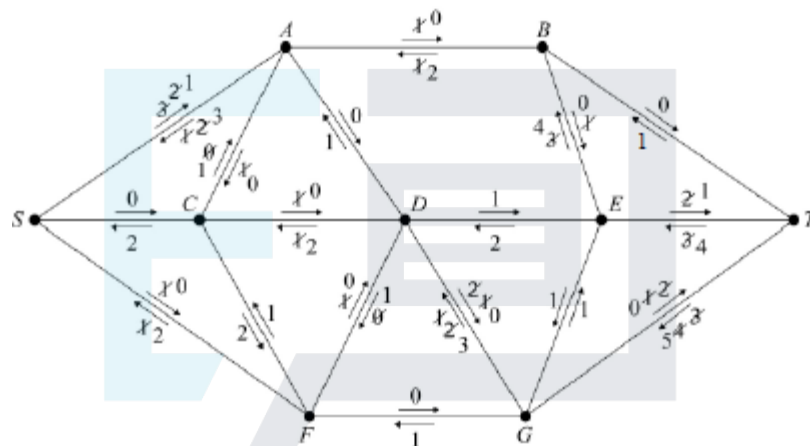
M1

all correct

DEG triangle may have different flows with implications to triangle GET.

A1

Network



1 path correctly augmented forward and backward
but must have earned M1 in part (b)(ii)

M1

network correct

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A1

4

(c) (i) Max flow = 37

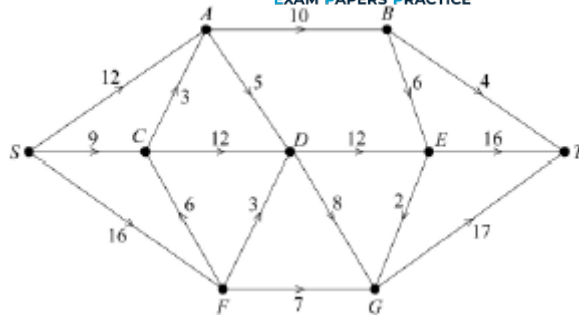
B1

1

(ii) Max flow

correct flow of 37

B2



condone 2 slips or omissions in flow of 37 **or** "correct"
feasible flow of 36
for SC1

2

- (d) Cut through AB, AD, CD, FD and FG
 $\{S, A, C, F\} \{B, D, E, G, T\}$

B1

1

[14]

Q5.

- (a) (i) $18 + (0 +) 10 + 3 + 5 \quad (= 36)$

B1

1

- (ii) 30, 32, 36 (missing cut values)
B1 each value correct

B3

3

- (iii) Max flow = 29

Award B0 E1 if their $\min^m$ cut is < 29

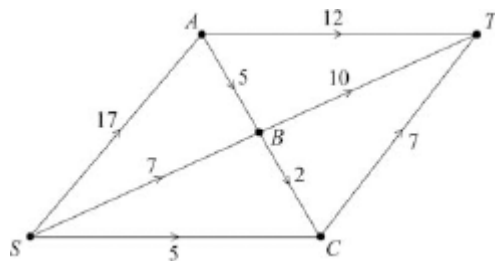
B1

because value of minimum cut is 29
and $\min^m$ value explained as max flow

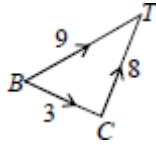
E1

2

- (iv)



may have



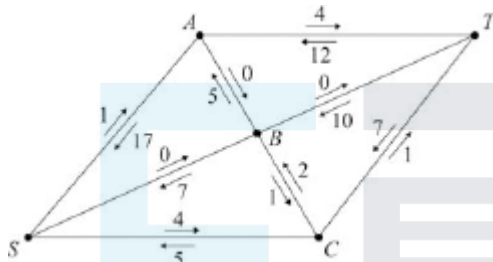
see alternative
solution on next
page

B1

cao

2

(b) (i)



potential flow (forward and back)
4 pairs 'correct' including SC and AT
ft their (a)(iv) provided $0 < \text{flow} < 30$

M1

all pairs correct (condone missing 0s)
ft their (a)(iv) if correct flow < 29

A1 ✓

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one correct flow in table

m1

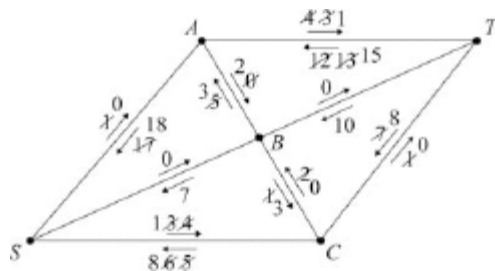


table
correct

SAT	1
SCT	1
SCBAT	2

If (a)(iv) flow < 29 then may score A1 for correct table
giving max flow of 33

A1

(see also the alternative solution)
modifying flows (forward and back)
1 flow correct ft their initial flow

m1

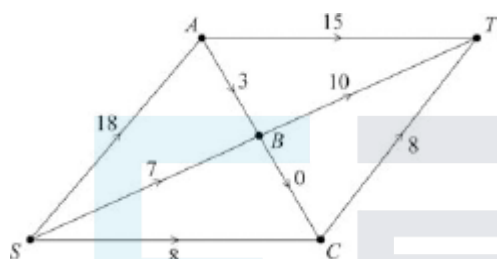
modified flows all correct, including all 0s (may score A1
from a correct flow < 29 seen in (a)(iv) if final flow correct)

A1

6

(ii) new max flow = 33

B1



6 flows correctly interpreted from their labelling procedure
provided M2 or M3 scored in (b)(i)
(may have AB 2, AT 16, BT 9 – see over)

M1

flow correct

SC B1 if flow of 33 shown correctly but not from correct
labelling procedure

A1

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3

[16]

Q6.

- (a) $SP \geq 12$
 $SQ \geq 10$
 $SR \geq 17$

S in correct place, (arrows) and capacities

B1

- $YT \geq 18$
 $ZT \geq 17$

T in correct place, (arrows) and capacities

B1

2

(b) $S P U Y T$ 10

B1

$S R V W Z T$ 8

B1

2

(c) (i) Initial flow forward and backward

PU 2 and 10 ; UY 0 and 10

B1

RV 0 and 8 ; VW 1 and 8 ; WZ 2 and 8

withhold one B1 if paths to S and T not updated

B1

2

(ii) Two correct routes and flows on Figure 2

$S P U Y T$	10	
$S R Y W Z T$	8	
$S P U X Y T$	2	
$S Q V U X Y T$	6	
$S R W X Z T$	5	
$S R W Z T$	2	

(other possibilities)

M1

Correct additional flows
Max flow = 33

A1

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Adjustment of at least 4 edges corresponding to flows
(forward and backward)

M1

Correct final flows forward and backward
(must score A1 for table)

edges UY , UX , WX and WZ will be saturated

$XY + XZ = 13$ in back flow

A1cso

4

(d) Cut with value 33 is through
 UY , UX , WX and WZ

B1

1

[11]

Q7.

- (a) Cut value = $40 + 27 + 0 + 24$
 = 91

B1

1

- (b) *ABDX* 16

B1

GFBX 18

B1

GHEX 20

B1

3

- (c) (i) One correct route with additional flow
 any feasible route and flow

M1

Another 2 routes and flows correct
 total flow at least 80

A1

All routes correct with total flow = 85

A1cso

Forward and backward flows on diagram
 (directions must be clear)

at least 8 edges with pairs of values 'correct'

M1

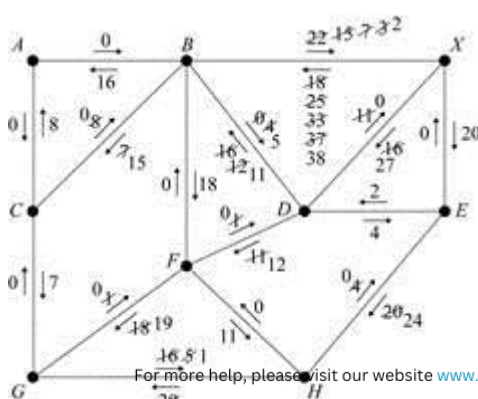
Augmenting flows
 correct

A1cso

5

Consider other possible correct flows

*Condone diagram as shown but really
 should have initial flows in DE, etc*



Route	Flow
<i>ABDX</i>	16
<i>GFBX</i>	18
<i>GHEX</i>	20

8

$GCBX$	7
$ACBX$	8
$GHFDX$	11
$GHEDBX$	4
$GFDBX$	1

(ii) Max flow = 85

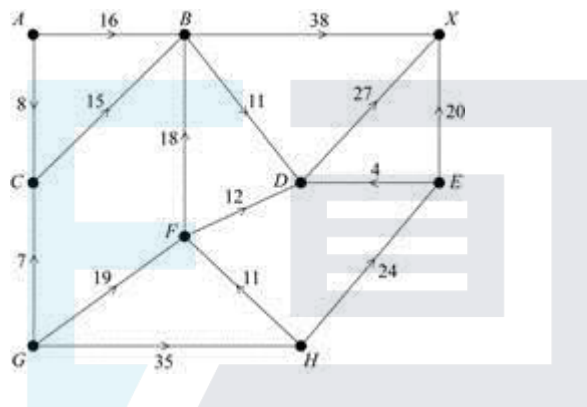
B1

Correct max flow

B1

2

Consider other possible correct flows



(d) Considering 'their' $AB+CB+FB - 45 = 4$ fewer

M1

$\Rightarrow$ max number = 81

EXAM PAPERS PRACTICE

A1cao

2

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[13]

Q8.

(a) (i) Value of cut = $38 + 25 + 0 + 0 + 34$

Must show correct addition

= 97

AG

B1

1

(ii) $\{S, A\}, \{B, C, T\}$ 65

B1

$\{S, B\}, \{A, C, T\}$ 57

B1

$\{S, B, C\}, \{A, T\}$ 72

B1

$\{S, A, B, C\}, \{T\}$ 56

B1

4

- (iii) Maximum flow = 53
ft their least cut value

B1ft

Minimum cut = Max flow

E1

2

- (iv) Their max flow on SA, SB or AT, CT

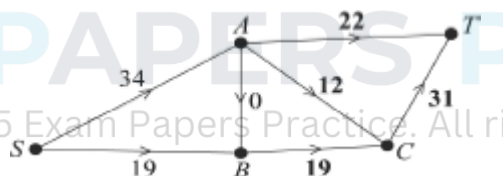
M1

All correct

A1

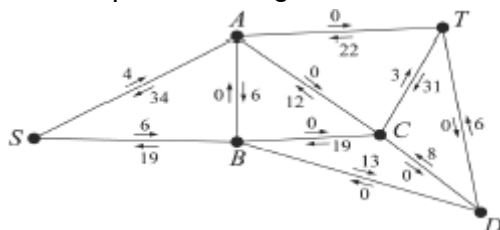
2

AT 22; AC 12; BC 19; CT 31
 $AB = x$; $AS = x + 34$; $SB = 19 - x$
 $0 \leq x \leq 4$



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- (b) (i) Initial flow on Figure 3
Forward potential and backward flow
Condone 2 slips, ft their Figure 2



M1

One correct augmented path in table and correct flow

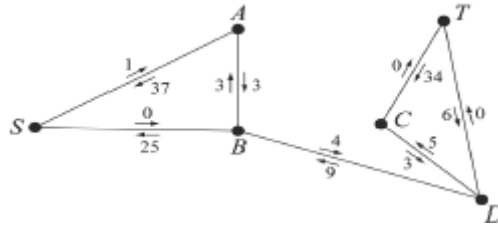
Path	Additional Flow
$SBDT$	6

M1

Table correct with total additional flow = 9

A1

Final network correct with evidence of labelling procedure used



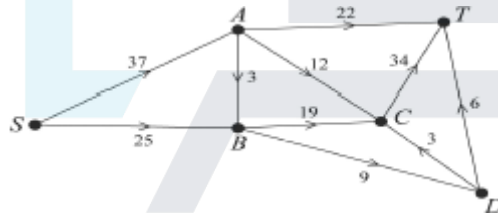
A1

4

(ii) New maximum flow = 62

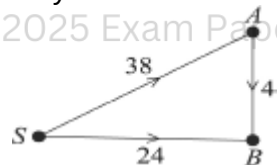
B1

Correct maximum flow on network



B1

May have



2

[15]

Q9.

(a) Value of cut = $10 + 10 + 15 - 4 - 1$
condone one slip if working shown

M1

= 30

A1

2

(b) BT 2, DE 3, ET 12

any 2 correct

B1

all correct

B1

2

- (c) (i) Initial flows forward and back or double
Arc with arrows (at **least 6 pairs correct**)
Condone pairs of values, (coordinates)
with single arrow

M1

all correct (condone pairs with single
arrow provided key indicated)

A1

2

(ii)

Path	Flow
SABT	2
SCDET	1
SACBT	1

(or SCBT instead of SACBT with flow 1
first correct path and flow

M1

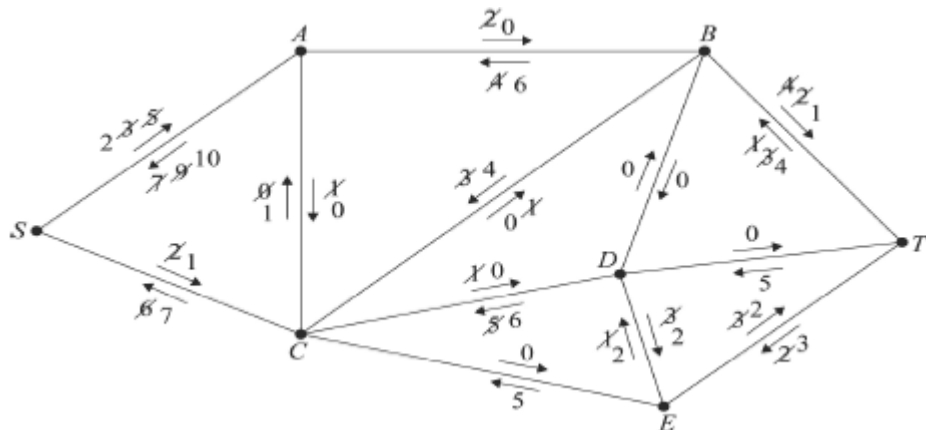
another correct path and flow

A1

all correct
(other possibilities also)

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A1



Must have forward and backward flows
augmenting flows (6 pairs correct)

M1

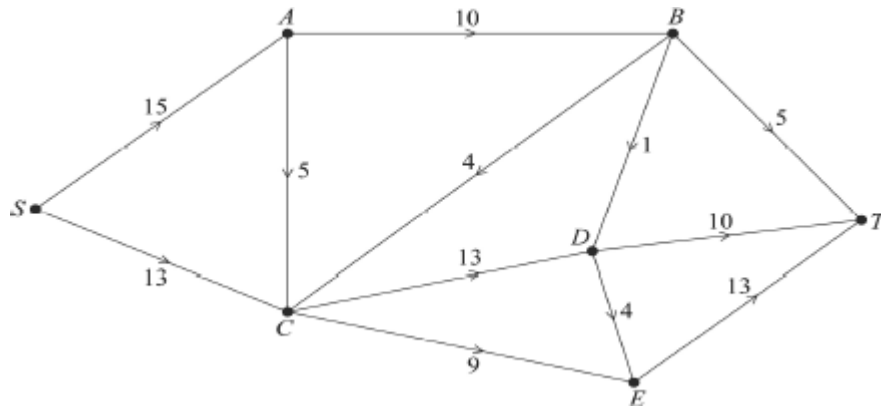
correct

A1

Alternative SA (3 & 9) SC (0&8)

5

(iii)



May have
SA (14), SC (14)
and AC (4) using
alternative
Maximum flow values

at least 8 correctly interpreted from their
Figure 3 but $24 < \text{their maxflow} < 29$

M1A1

2

- (d) Cut through AB, CB, CD and CE
May use $\{S, A, C\} / \{B, D, E, T\}$

But must have total flow of 28 in their
network (condone one slip)

B1B1

1

[14]

Q10.

- (a) Arrival gates are U and R

B1

1

- (b) Cut value = $45 + 53 + 20 + 37 + 0$
= 155

B1

1

- (c) Max flow along UTSP is 17

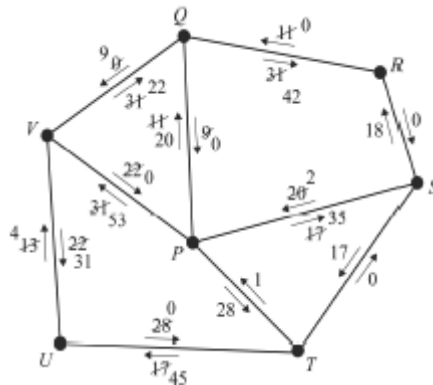
B1

and along $RQVP$ is 31

B1

2

(d) (i)



Route	Value of Flow
$UTSP$	17
$RQVP$	31
RSP	18
RQP	11
UTP	28
UVP	22
$UVQP$	9

Initial flows along $UTSP$ and $RQVP$ with potential increases and decreases

B1

Table: first route and correct flow

After $UTSP$ and $RQVP$

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M1

Another route and flow

A1

Table correct

A1

Network: attempt to use labelling procedure with forward/backward flows

M1

All diagram correct

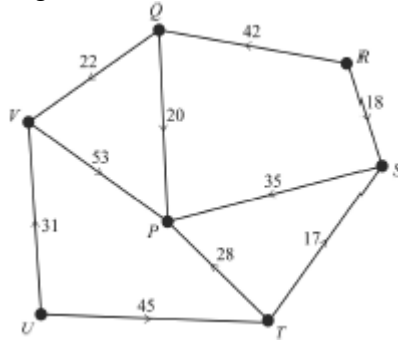
A1

6

(ii) Maximum flow = 136

B1

Figure 2 correct:



Other possible answers

B1

2

(e) Rate reduced by 3
"their" maximum flow – 3

M1

New maximum is 133

A1

2

[14]

Q11.

(a) Value of cut = $30 + 10 + 12 + 20$

M1

= 52

Full marks for correct answers without working

A1

2

(b) $AE = 9$;

B1

$EF = 5$;

B1

$FG = 4$

B1

(c) (i) Attempt at forward and backward flows

SA 2 & 4; AB 1 & 3; BT 1 & 3
SD 3 & 1; DA 0 & 3; AE 0 & 3
BE 0 & 7; DE 2 & 0; ET 1 & 3
FD 2 & 1; EF 5 & 1; EG 1 & 5
FG 1 & 2; GT 3 & 0

At least 5 pairs correct

M1

10 pairs correct

A1

all correct

A1

3

(ii) First flow augmenting path and correct flow on table

M1

Table correct

A1

Adjusting flows – forward and back

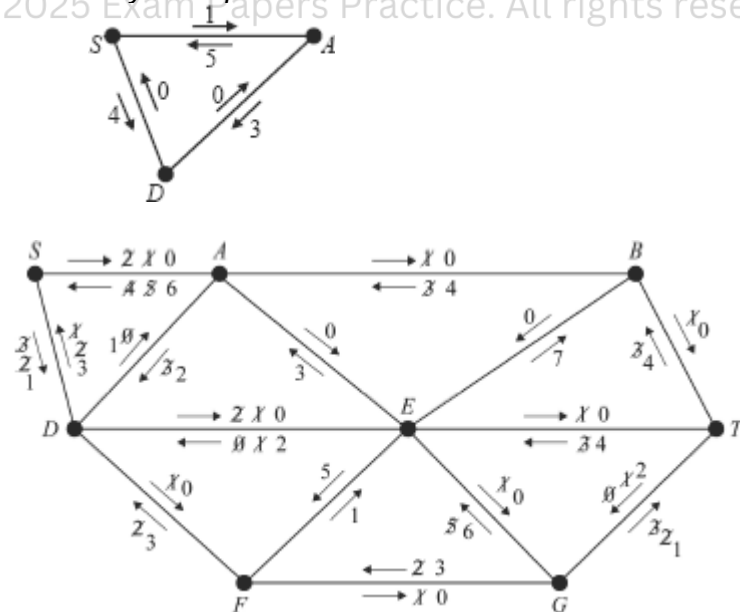
M1

Correct

A1

4

May end up with



Path Extra flow |

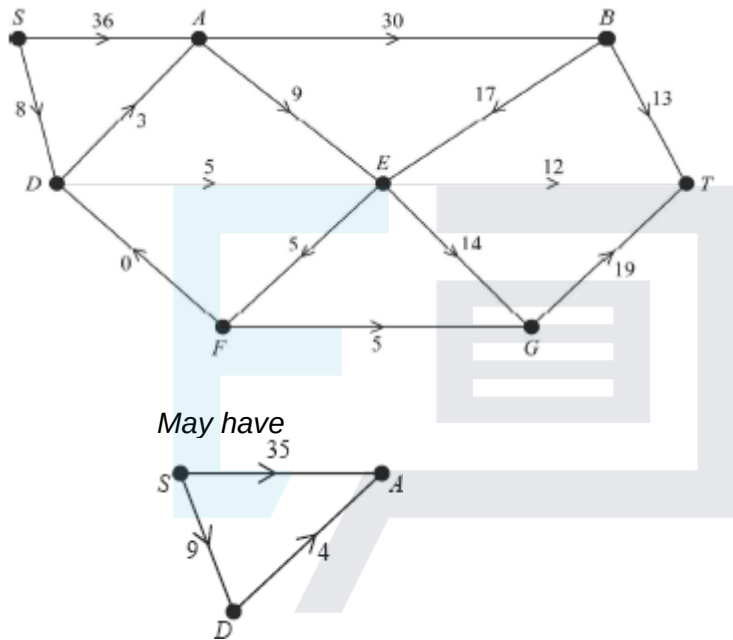
SABT	1	Or SDET
SADET	1	
SDFGT	1	
SDEGT	1	

- (d) Max flow of 44 shown on figure 3
up to 2 slips

M1

all correct

A1



- (e) Cut through their saturated arcs

M1

Cut passes through AB , AE , DE and DF
Or BT , ET , EG , and FG

A1

2

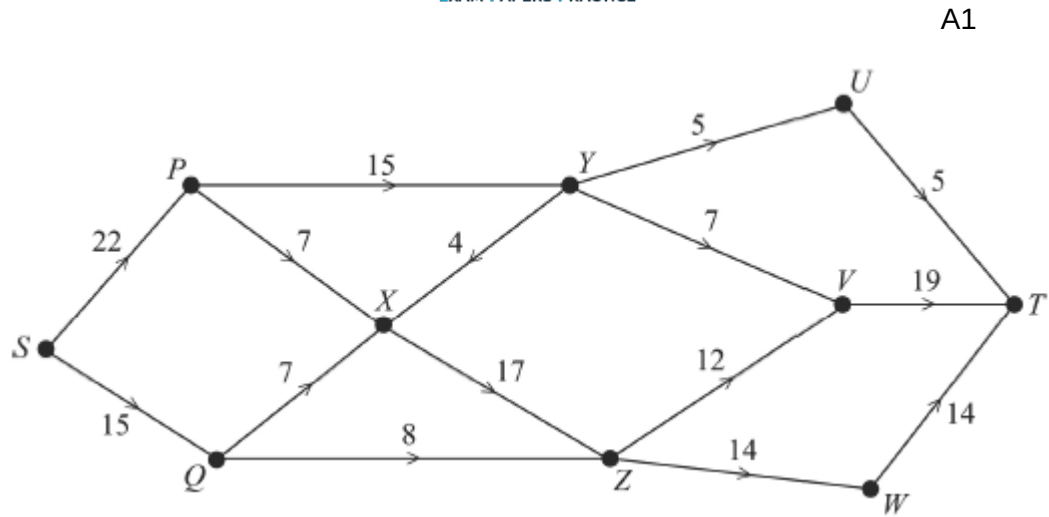
[16]

Q12.

- (a) Correct position of S and T

M1

Values on edges SP , SQ , UT , VT and WT



A1

2

- (b) (i) Cut C has value 40
 $15 + 0 + 17 + 8$

B1

1

- (ii) Max flow ≤ 40

E1

1

- (c)

Route	Flow
$SQZWT$	8
$SPYXZVT$	4

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B1B1

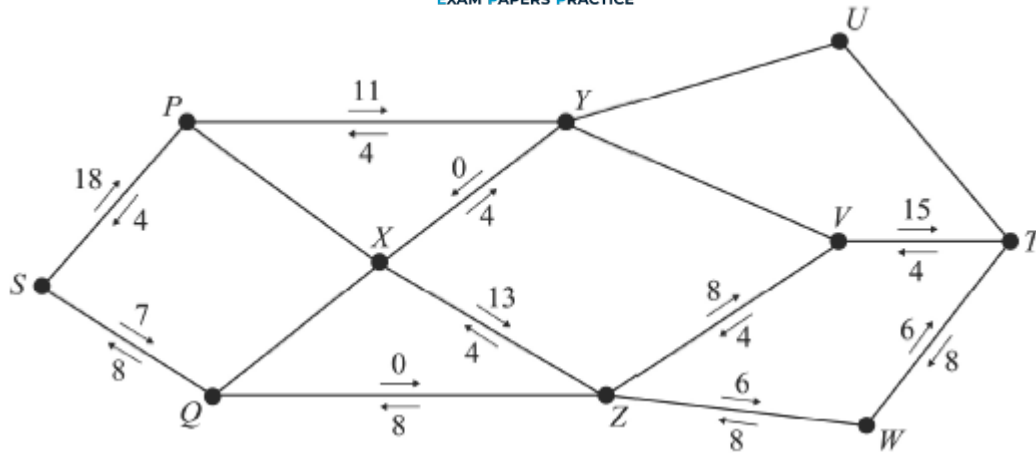
2

- (d) (i) 3 forward and backward flows correct

M1

All initial values correct on edges below

A1



2

(ii)

Route	Flow
SQZWT	8
SPYXZVT	4
SPYUT	5
SPYVT	6
SPXZVT	7
SQXZWT	6
SQXYVT	1

(Many different possibilities)
2 or more correct flows in table

M1

Table correct (adding to 37)

A1

At least 2 flows augmented on diagram

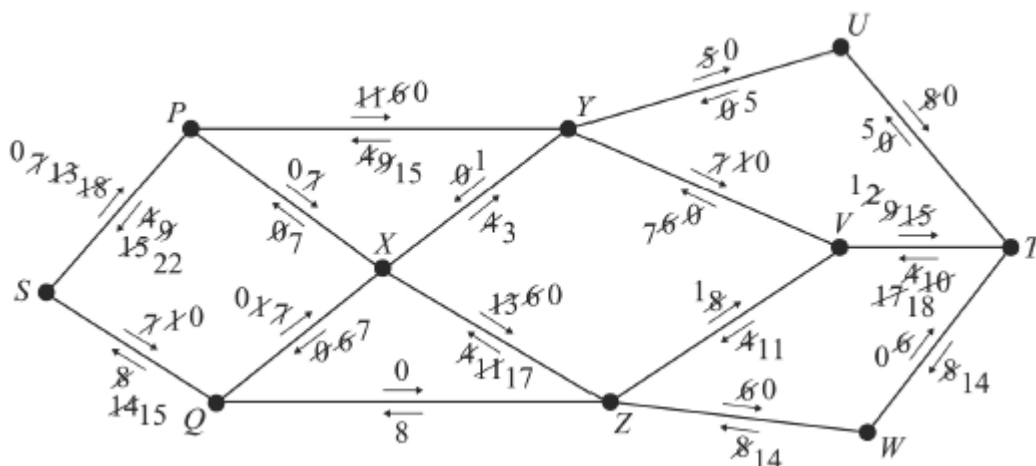
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M1

Correct forward and backward final flows

A1

4



Other possibility for ZV, VT, ZW and WT

- (e) Flow from Y to X is 3

B1

1

[13]

Q13.

- (a) (i) $15 + 0 + 14 + 7 + 9 = 45$

B1

1

- (ii) Maximum flow ≤ 45
 $\leq$ their value or < 45

correct

M1

A1

2

- (b) *SABT* flow 10
SDET flow 14
SFT flow 9
one correct

two more correct

B1

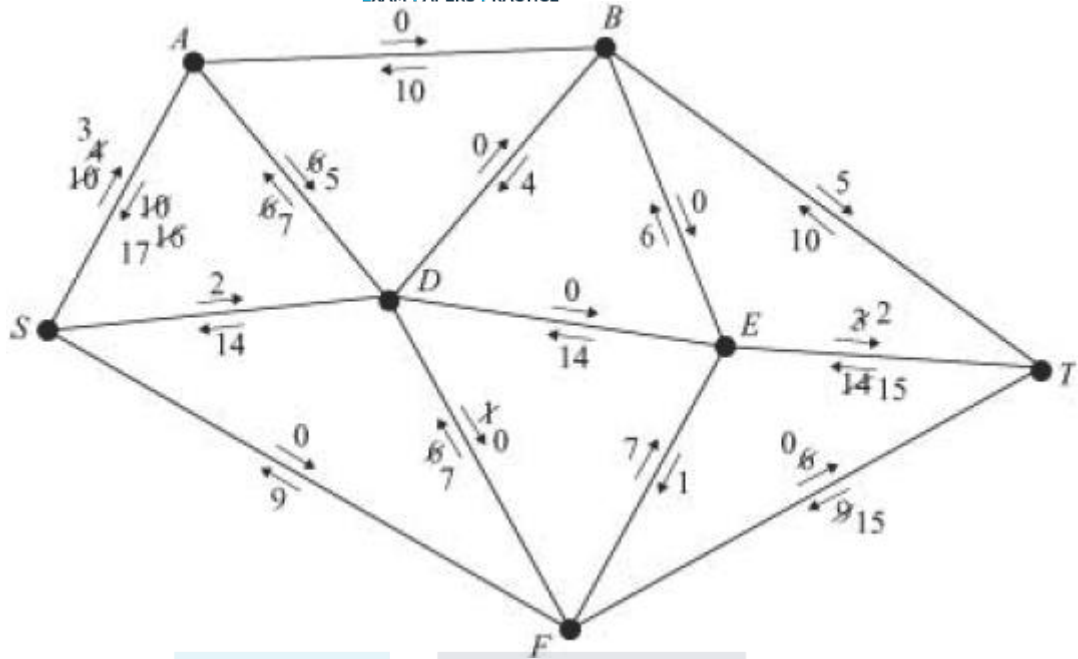
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B1

(may appear in table below)

2

- (c) (i)



Additional route

M1

with correct flow

A1

one more correct route and flow

A1

table complete

correct total flow of 40

A1

correct use of potential and used flows

on network (may use double edges)

M1

values correctly updated

strict

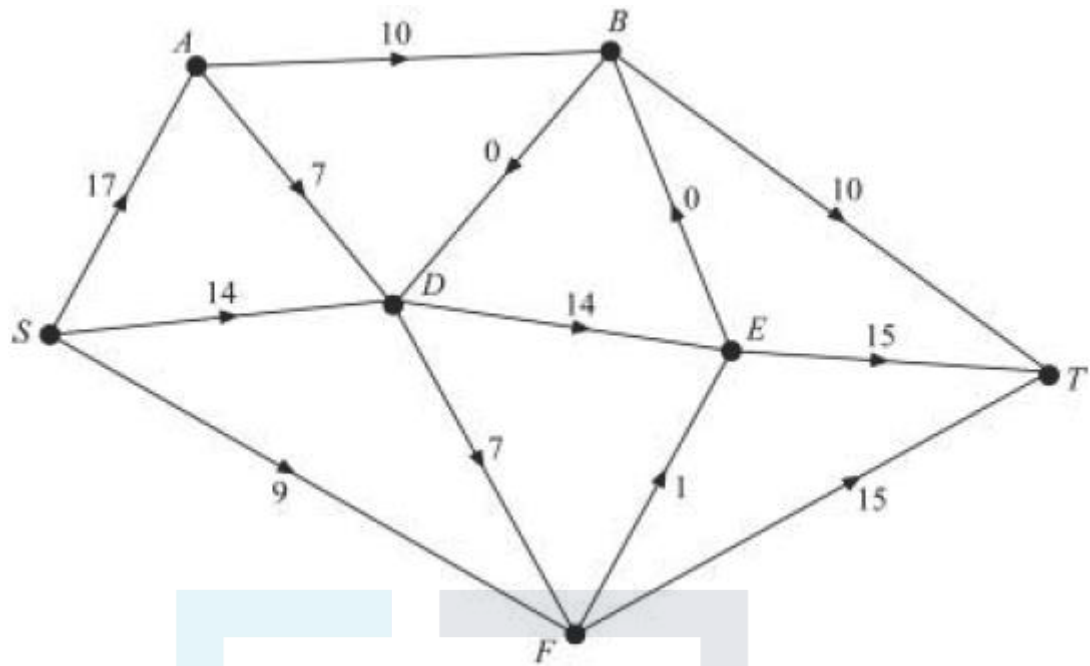
A1

6

Route	Flow
SABT	10
SDET	14
SFT	9
SADFT	6
SADFET	1

several possibilities

(ii)



Maximum flow = 40

Network showing flow of 40

B1

B1

2

(iii) Cut through saturated arcs
AB, BD, DE, DF, SF

M1

Minimum cut shown to be 40 with statement
linking to maximum flow

A1

2

[15]

Q14.

(a) (i) $5 + 8 + 16 - 3 = 26$

B1

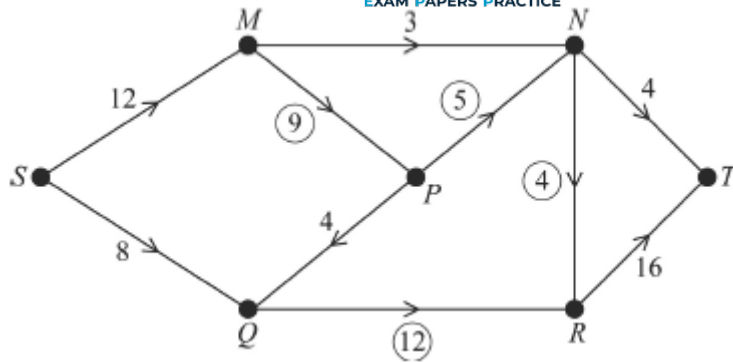
1

(ii) Max flow ≤ 26

E1ft

1

(b)



$MP - 9$

$PN - 5$

$NR - 4$

$QR - 12$

B1

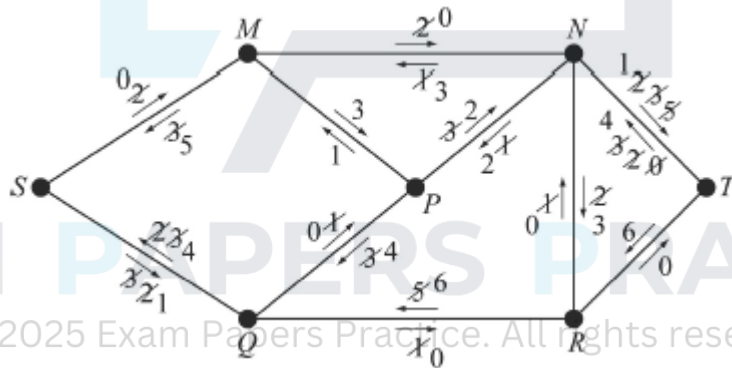
B1

B1

B1

4

(c) (i)



*initial flow – forward and backward
6 pairs correct*

M1

correct

$OM\ 2\ \&\ 3;$ $MN\ 2\ \&\ 1$

$NT\ 5\ \&\ 0;$ $MP\ 3\ \&\ 1$

$SQ\ 3\ \&\ 2;$ $PQ\ 3\ \&\ 1$

$PN\ 3\ \&\ 1;$ $QR\ 1\ \&\ 5$

$NR\ 2\ \&\ 1;$ $RT\ 0\ \&\ 6$

A1

2

(ii) Adjusting flows on network

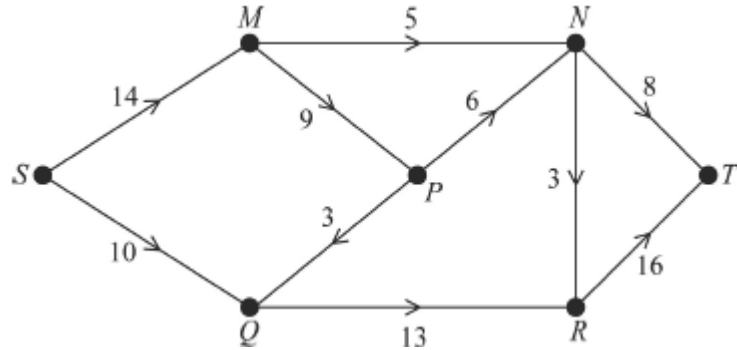
M1A1

Path	Flow
SMNT	2
SQPNT	1
SQRNT	1

First correct path and flow B1
 Second correct B1
 Rest B1

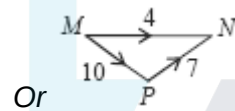
5

(iii)

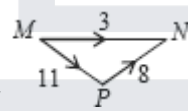


6 flows correct

all correct



or



M1

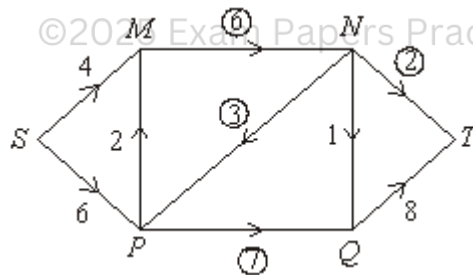
A1

2

[15]

Q15.

(a)



MN

NT

PQ

NP

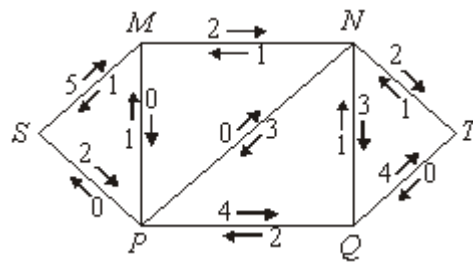
B1

B1

B1

B1

(b) (i)



initial flow indicated as surplus forward and backward flows

M1

use of flow augmentation

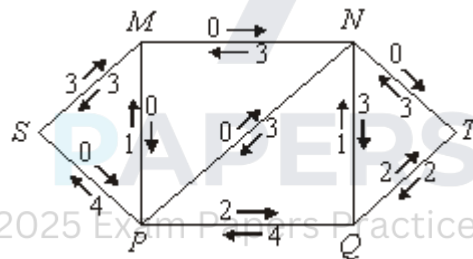
M1

e.g. SMNT 2
one flow correctly identified

A1

SPQT 2
all possible flows correct

A1



amending flows (dep on first M1)

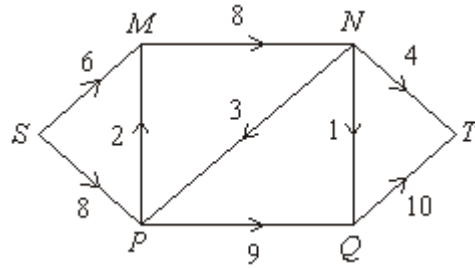
M1

final situation with saturation
at M and P

A1

(ii) Max flow – 14

B1



B1

2

- (c) Cut through 2 of their saturated arcs
cut on **original** network

M1

described or drawn

A1

2

[14]

Q16.

- (a) D

B1

1

- (b) $(17 + 25 + 35 + 13 + 12 + 13 = 115)$

B1

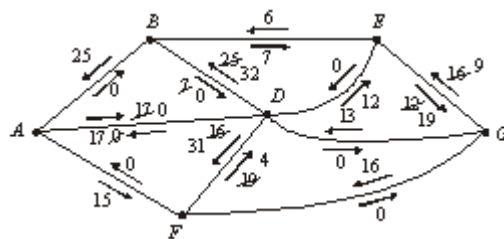
1

- (c) $ABD_{\max} = 25$; $GED_{\max} = 12$

B1B1

2

- (d) (i)



Forward and backward flows

M1

Adjusting flows on diagram

M1

Routes and flows in chart

M1

One correct other than ABD, GED

A1

Another correct

A1

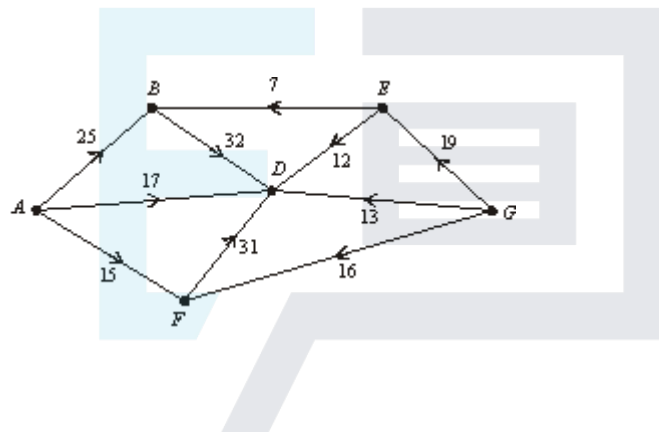
Route	ABD	GED	GFD	GD	AD	AFD	GEDB
Flow	25	12	16	13	17	15	7

All correct

A1

6

- (ii) Total = 105
Max flow



B1

B1

2

- (iii) Cut through AF, AD, BD, DE, DG, and GF
Through 3 saturated arcs (fairly generous)

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Correct

A1

2

- (e) Reduce max flow by their EG
changing 19 to 15
Reduce by 4 since everywhere else saturated

M1

⇒ New max = 101
Correct answer ⇒ 2 marks

A1

2

[16]

Q17.

- (a) (i) $9 + 7 + 0 + 9 + 13 = 38$

B1

1

- (ii) Maximum flow is less than or equal to 38
< their value of cut

M1

$$\leq 38$$

M0 for "equals" their cut

A1ft

2

- (b) *SUYWT* flow of 9

B1

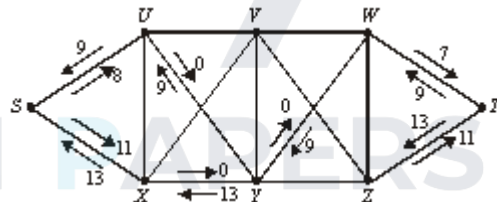
SXYZT flow of 13

B1

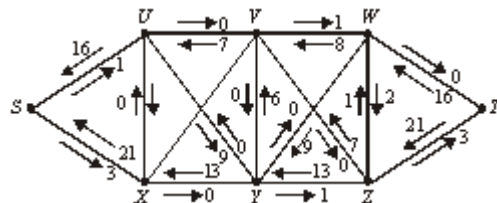
2

- (c) (i) Indicating flows from (b) on network
Preferably as backward flows

M1



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 with augmentation leading to something such as



Route	Flow
<i>SUYWT</i>	9
<i>SXYZT</i>	13
<i>SUVWT</i>	7

SXVZT	7
SXVWZT	1

Flow augmentation (many possibilities)

$SUVWT \quad 7$

M1A1

$SXVZT \quad 7$

m1A1

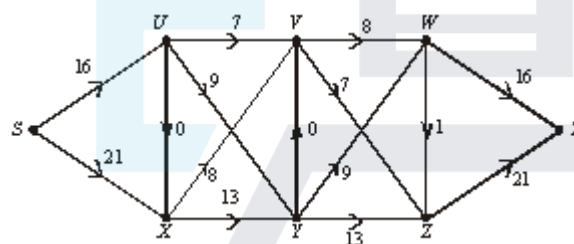
$SXVWZT \quad 1$

A1

6

- (ii) Network showing maximum flow
Several possibilities

B1



Maximum flow is 37

Or $\{S, U, X / V, W, Y, Z, T\}$

B1

2

- (iii) Attempt to find cut through saturated arcs

M1

Cut through UV, UY, XV, XY

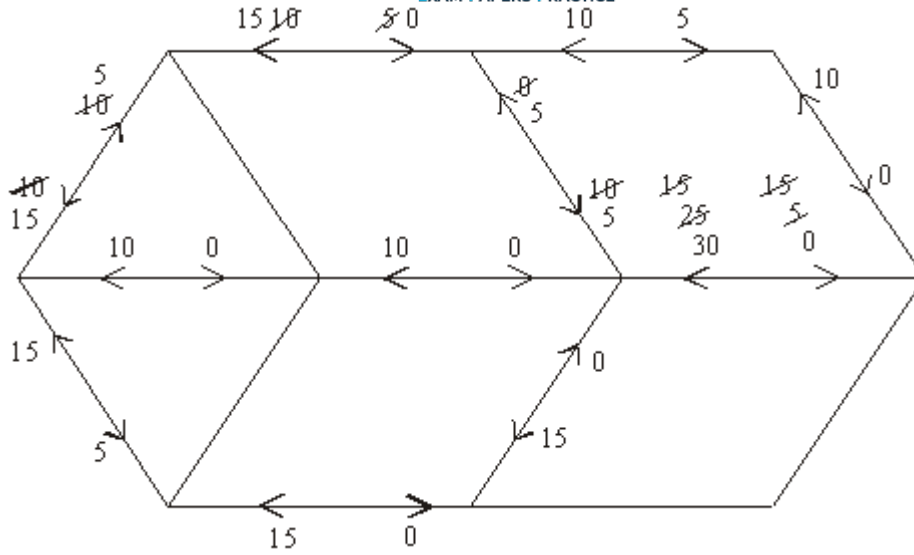
A1

2

[15]

Q18.

(a)



SCA

first flow

second flow

$\therefore$ maximum flow = 40

M1

A1

A1

B1

4

(b) Cut BE, DG, CF

B1

1

(c) Out of A – max 50

B1

$\therefore$ from A must be road

Same as J

B1

$\therefore$ join AJ direct

2

[7]

Q19.

(a) A, B, C, D

B1

1

(b) /

B1

1

(c) $20 + 20 + 10 + 10 + 40 + 50 + 30$

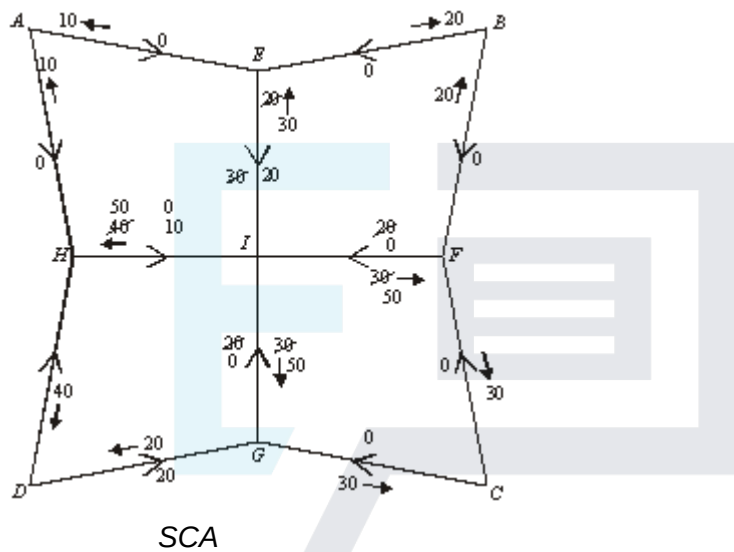
M1

$= 180$

A1

2

(d)



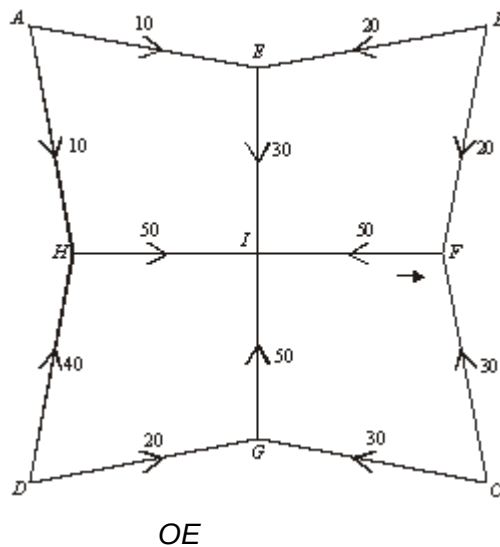
1 complete flow

M1

M1

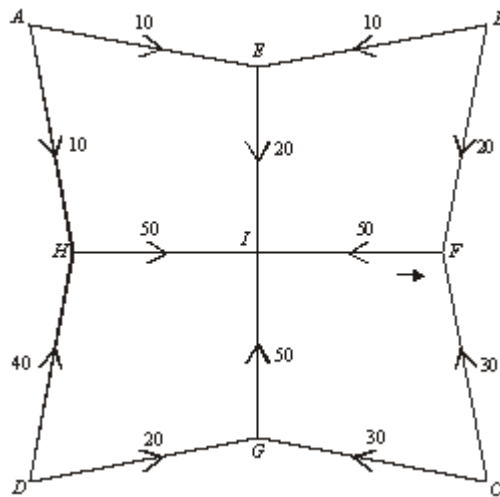
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Final:



A1

(e)



Their max – 10

Max flow = 170

CAO

M1

A1

2

[11]

Q20.

(a) (i) Source A, C

B1

1

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(ii) Sink G, I

B1

1

(b) (i) Flow = 100

B1

1

(ii) $x = 14$

B1

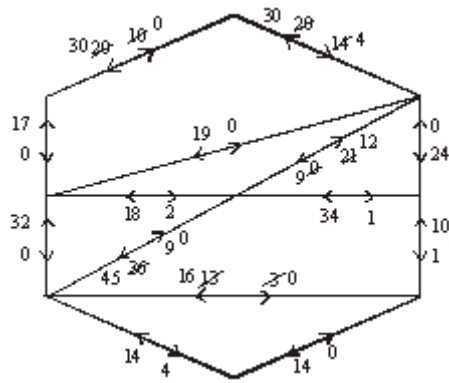
1

(iii) $y = 34$

B1

1

(c) Flow = 122



Initial diagram

SCA

1 flow; OE

2nd flow

All correct; OE

M1

M1

A1

M1

A1

6

[11]

Q21.

(a) $C_1 = 42$

B1

$$C_2 = 10 + 8 + 14 + 6 + 10$$

M1

$$= 48$$

A1

3

(b)



M1*

M1*

M1*

A1*

A1F*

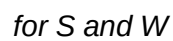
A1*

B1

7

[10]

(a)



2

M1

for five arcs correct ($\geq$)

A1

(b) $SADYW = 10$

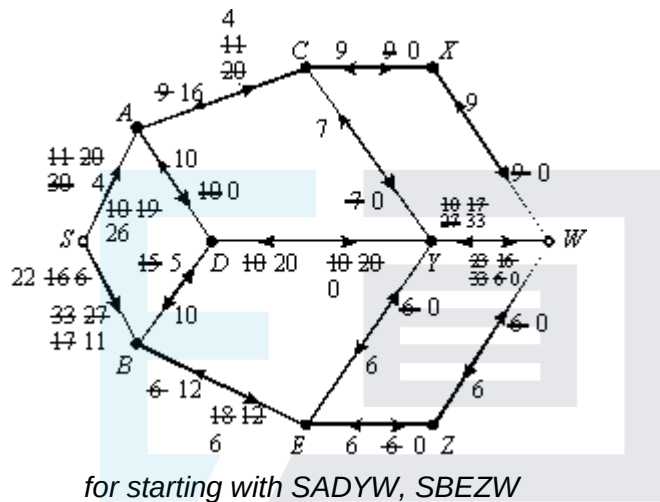
B1

$SBEZW = 6$

B1

2

(c) (i)



M1

SCA

M1

for AC

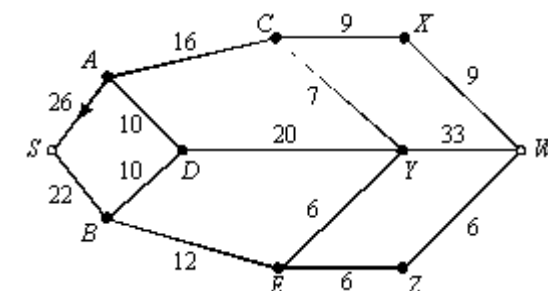
A1

for SA

A1

for SB

A1



OE

B1

6

(ii) Maximum flow = 48

M1

Minimum cut CX, YW, EZ
 (9) (33) (6)
 = 48

A1

2

[12]

Q23.

(a) (i) Flow = 25

E1

(ii) $15 \leq \text{Flow} \leq 25$

E1

2

(b) At F
 Min in = 7

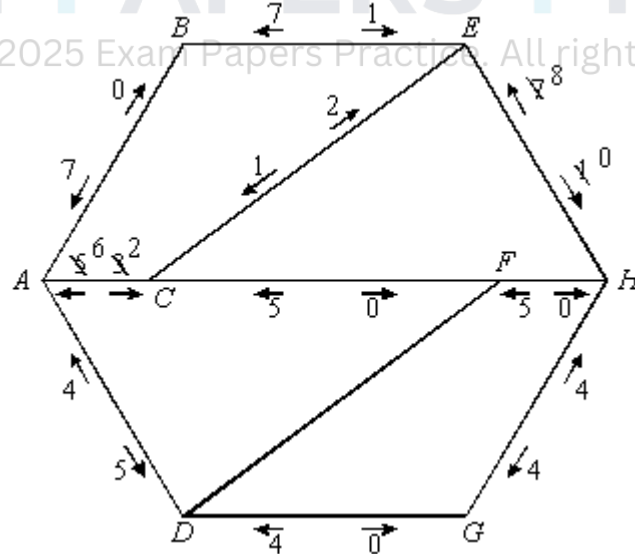
M1

Max out = 5

A1

2

(c) (i)



SCA

M1

1 complete flow

M1 A1

2nd complete flow

A1

All correct

A1

5

(ii) Any cut = 17

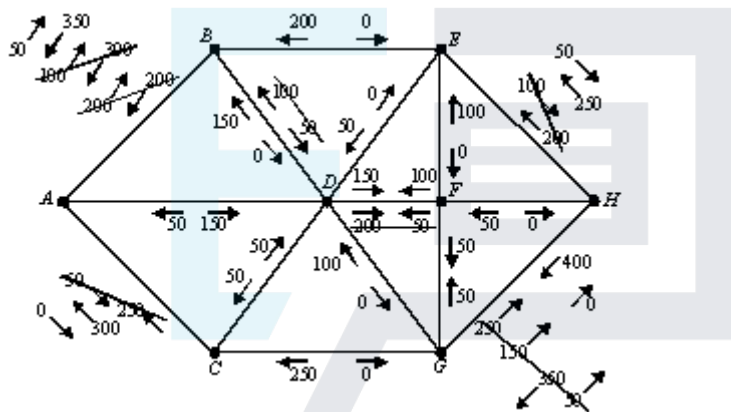
B1

1

[10]

Q24.

(a)



AB or AC started

M1

both initial correct

A1

BE + CG

A1

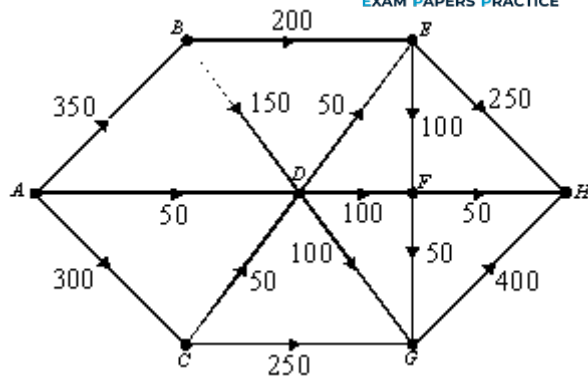
GH, EH

A1

all correct
o.e.

A1

Final solution



o.e.

B1

6

- (b) Minimum BE, DE, FE, FH, GH
 (200) (5) (0) (5) (400)
condone omitting FE

= 700

M1

A1

2

[8]

Q25.

- (a) **SC, SB, AT**

M1 A1

2

- (b) (i) **SCT** 1

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M1 A1

2

- (ii) **$SABCT$** 2

M1 A1

2

- (c) any flow $\leq$ any cut
 so any flow ≤ 16

B1

1

- (d) Choose **AT** .

B1

The arc must be in the minimum cut in

M1

(a). Also it must be in $\{AT, BT, CT\}$
which is a cut of 17.

A1

3

[10]

Q26.

- (a) e.g. $SABT$ 4
 $SACBT$ 2

A1

$SCDT$ 2

A1

$SEDFT$ 3

A1

$SEFT$ 3 (total 14)

*Any two paths
Remaining paths to
make total 14
(or 2 for cao)*

A1

5

- (b) EF, ED, CD, BT
(3 + 3 + 2 + 6 = 14)

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M1 A1

2

- (c) (i) Flow in $SE \leq$ flow out at $E \leq 3 + 3$

B1

1

- (ii) Flow into $C =$ flow out of $C \leq 3 + 2 = 5$

M1

So if AC has a flow of 4 then SC has a
flow of at most 1.

A1

2

- (iii) If AC is saturated the maximum flows in
 SA, SC and SE are 6 (its capacity), 1 (by

part (ii)) and 6 (by part (i)). So if AC is saturated the maximum flow out of S is $6 + 6 + 1 = 13$ which is less than the unrestricted maximum flow.

M1

A1

2

[12]

Q27.

(a) (i) 30

(ii) 25

B1 B1

2

(b) $YZ = 5$ $XZ = 0$

$WX = 5$ $SW = 15$

M1 A1

A1

3

(c) $SYZT$

M1 A1

$SYWXT$

M1 A1

4

(d) Total flow in (c) = $15 + 5 + 5 = 25$

M1

Max flow $\leq$ any cut, so no flow can exceed the 25 in (a)

(ii)

A1

2

[11]

Q28.

(a) Vertices S and T

M1

Arcs SS_1 , SS_2 , T_1T , T_2T and T_3T

Capacities 18, 15, 10, 13, 12 (or more) respectively

A1
2

(b) (i) $8 + 2 + 5 + 4 + 12 = 31$

B1
1

(ii) $AB \ AC \ DC \ DE$

M1

(or $AB \ CB \ CT_2 \ CE \ DE$)

A1
2

(c) e.g. $SS_1ABT_1T: 8$

M1A1

$SS_1DET_3T: 7$

A1

$SS_2DCET_3T: 5$

A1

$SS_2DACT_2T: 5$

A1

$SS_1ACBT_2T: 2$

$SS_1DCET_2T: 3$

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A1
6

(d) All flows $\leq$ all cuts

M1

So, by (b)(ii), all flows ≤ 30 .
Hence the flow of 30 is maximum possible.

A1
2

(e) e.g. For T_1 to get 10 BT_1 will have a flow of 10.

M1

Then, looking at vertex B , max inflow = 10.
Hence BT_2 has 0 flow.

A1

So maximum arriving at T_2 is from CT_2 and ET_2 with a total capacity of 9.

A1

3

[16]

Q29.

- (a) Vertex S , arcs SS_1 and SS_2 with

M1

capacities at least 10 and 11 respectively.

A1

2

- (b) e.g. SS_1ABT 4

M1

SS_1CDT 5

A1

SS_2CDBT 3

A1

SS_2EFT 6

A1

SS_2CEFT 2

A1

5

- (c) S_1A, CD, EF

M1

Is a cut of 20.

A1

So all flows ≤ 20

A1

3

[10]

Q30.

(a) e.g. *SCT* 2

M1

SCBT 1

A1

SACBT 2

A1

SABT 2
giving a flow of 7

A1

4

(b) *AB, CB, CT* is a cut of 7

M1

(So the flow of 7 is a maximum)

A1

2

[6]

Q31.

(a) e.g. 300

M1

SABT 100, *SADCBT* 50

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A1 A1

SADCEBT 100, *SADCBT* 50

A1 A1

Flow of 650

Max of 3 for final flow only

B1

6

(b) *DE, CE, CB, AB*

M1 A1

2

(c) Can augment *SDCT* by 50

B1

New network still has a cut of 700 (*AB,AC,DC,DE*)

M1 A1

3

[11]



EXAM PAPERS PRACTICE

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Examiner reports

Q2.

Most students answered part (a) correctly. Although there were many correct solutions for part (b)(i), a significant number of students failed to give a convincing solution on their diagram. It is essential that students use 'backward' arrows to show existing flows and 'forward' arrows to show potential flows. There was a significant number of students who scored 5 marks in part (b)(i) but failed to score both marks in part (b)(ii). Part (c) was answered poorly. Students were asked to list the edges of their cut; many simply drew a cut on their diagram.

Q3.

Although many students answered this part correctly, there was a significant number who included values on their edges that corresponded to the initial flow. In part (b), a large number of students started from a 'zero-flow' and then augmented to arrive at the given flow and then stopped. In (b)(i), although there were many correct solutions, a significant number of students failed to give a convincing solution on their diagram. It is essential that students use 'backward' arrows to show existing flows and 'forward' arrows to show potential flows. There were a significant number of students who scored 5 marks in part (b)(i) but failed to score both marks part (ii). Part (c) was poorly answered. Students were asked to list the edges of their cut, many simply drew a cut on their diagram.

Q4.

Part (a) It was surprising to see a large number of able candidates being unable to calculate the correct value of the cut, with many confusing upper and lower capacities in their calculation.

Part (b)(i) Few candidates had trouble in finding the flows along DE and FG .

Part (b)(ii) It was pleasing to see that many more candidates this year were aware of how to show the potential increases and decreases along each edge. Those who did not do this lost both marks here and two marks in the next part of the question.

Part (b)(iii) It was good to see candidates trying to set out their solution in a logical manner and once again the diagram and table clearly helped. These candidates used the table effectively to show what new flows had been introduced and modified both the forward and backward flows on their network. Marks were awarded for the initial flow and it is very difficult to credit candidates for their original values if they have been obliterated during augmentation; candidates are advised to show the initial flow in ink and then to augment their flow using pencil.

Part (c) Those who stated that the maximum flow was 36 were given some credit for illustrating a flow of 36 on Figure 3. Those with the correct maximum flow were usually able to show a correct flow on the network; those who made a couple of slips were again given some credit for their work.

Part (d) Many candidates were able to find the correct cut with capacity 37, but quite a few who had the correct maximum flow were deceived into using their Figure 3 on which every cut had value 37, rather than looking at the initial network with upper and lower capacities.

Q5.

- (a) (i) Many students were able to show the correct working to give the value of the cut.
- (ii) Far fewer were successful in finding the three values of the remaining cuts in the table.
- (iii) Some thought that the “maximum” cut was needed to give the maximum flow and others wrote “maximum flow $\leq$ minimum cut”. The 29 value had been given to help students and this proved valuable for those who had been unable to find the 3 correct values in the table.
- (iv) Those with a maximum flow of 29 were usually able to draw correct flows along the given edges, although some gave at least one value greater than the capacities on the original network.
- (b) Some missed the vital point of supplementing the flows by 4 along SC and AT and made no further progress. It was good to see many students trying to set out their solution in a logical manner and once again the diagram and table clearly helped. There are still many students unfamiliar with the “labelling procedure” who fail to show both potential forward **and** backward flows on their network. Students are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit students for their original values if they have been obliterated during augmentation.

Some credit was given to students who had used flow augmentation correctly but had failed to find a maximum flow equal to 33, if they interpreted their flows correctly on the final diagram.

Q6.

In part (a) most candidates understood how to add a super source and super sink to the network and inserted correct weights on the edges.

In (b) again, few candidates had trouble in finding the maximum flows along the given edges. It was not clear whether some had misread the second route when they gave the flow as 9 instead of 8. For this reason, this error was not penalised heavily in the subsequent work if they eventually found the correct maximum flow through the network.

In part (c) it was good to see candidates trying to set out their solution in a logical manner and once again the diagram and table clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit candidates for their original values if they have been obliterated during augmentation.

In (d) those who had a correct maximum flow in part (c) were usually able to find the correct cut with value equal to the maximum flow.

Q7.

- (a) Many candidates failed to obtain the correct value of the cut, usually as a result of subtracting 6 instead of adding 0.

- (b) Again, few had trouble finding the maximum flows along the given edges.
- (c) It was good to see candidates trying to set out their solution in a logical manner and once again the diagram and table clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit candidates for their original values if they have been obliterated during augmentation. This was a good discriminator and tested that candidates understood the labelling procedure. Only the best candidates realised the need to augment a flow in the direction from *D* to *B*; not many obtained a flow greater than 80 and very few achieved the correct flow of 85 with a fully augmented correct diagram.
- (d) Most candidates realised that they needed to reduce their total flow by 4 because of the vertex restriction, but only the best candidates obtained the correct maximum number that could move through the network during the fire drill.

Q8.

Usually after crossing out one or two additional figures, in part (a)(i), most candidates managed to show that the cut had value 97. This(ii) had been intended as a help to find the values of the other cuts in part (a)(ii), but only the best candidates managed to find the 4 correct values in the table. In part (a)(iii), many candidates made statements involving inequalities instead of saying that the maximum flow was equal to the minimum cut. The cut with value 53 had already been provided in the table but several candidates ignored this value.

In part (b)(i), it was good to see candidates trying to set out their solution in a logical manner and once again the insert clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network; however, some candidates started from a flow of zero rather than using their values on Figure 2. Candidates are advised to use the table to show what additional flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. In part (b)(ii), those who had the correct maximum flow and who had used the labelling procedure correctly usually had no trouble in showing this maximum flow on the final diagram.

Q9.

Part (a) Most candidates seemed unaware of how to calculate the value of the cut correctly because of the upper and lower capacities. Some subtracted 9 instead of 5 from the sum of the other upper capacities; others merely added a set of positive numbers.

Part (b) In contrast, almost everyone scored full marks for finding the value of each of the missing flows along the given edges.

Part (c)(i) Some candidates still seemed unaware of how to represent the potential increases and decreases **from the initial feasible flow**. It requires forward and backward arrows or a duplicate edge; one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows so that any adjustments can be shown in pencil so as not to obliterate the initial flow figures.

Part (c)(ii) A table was provided so that a flow of 2 along *SABT*, for example, could be listed in the table. The potential forward and backward flows along *SA*, *AB* and *BT* could

then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. If candidates obliterate their values of the potential increases and decreases from part(i) then they risk losing marks for that part. The majority of candidates found only 2 augmenting paths and were not able to find the correct maximum flow. Those who were not using the labelling procedure correctly sometimes obtained a flow that was impossible; for instance a maximum flow of 29 was seen quite often. It should be noted that candidates are not expected to do flow augmentation with negative flows.

Part (c)(iii) Some credit was given for transferring values from Figure 1 to Figure 2 even if the maximum flow had not been achieved; but full marks could only be scored for a correct flow of 28 through the network.

Part (d) Very few candidates realised the need to consider their saturated arcs in order to obtain the correct cut. Many wrongly considered their flow on Figure 2, where obviously every cut has a value of 28, and consequently wrote down cuts such as the one through *BT*, *DT* and *ET* as their final answer. That is why the question suggested that candidates “may wish to show the cut on the network above” since that network contained all upper and lower capacities.

Q10.

In part (a) most candidates identified the arrival gates and scored an easy mark.

In part (b) quite a few candidates made errors in calculating the value of the cut, with common wrong answers being 138 and 172.

In part (c) the values of the initial maximum flows were usually correct.

In part (d) it was good to see candidates trying to set out their solution in a logical manner and once again the insert clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made.

For part (d)(ii) the incorrect answer of 127 was seen far more often than the correct value for the maximum flow, since candidates seemed unaware of the need to introduce an additional flow of 9 by actually reducing the flow along some edges that were previously at full capacity.

In part (e) those who had the correct maximum flow in part (d) were usually successful in finding a new maximum flow and those who did not have a correct answer were given some credit for reducing their previous flow by 3 or explaining how the extra 3 passengers could be accommodated in some way.

Q11.

In part (a), most candidates seemed unaware of how to calculate the value of the cut correctly. Many subtracted 17 instead of 10 from the sum of the other upper capacities.

In contrast, in part (b), almost everyone scored full marks for finding the value of each of the missing flows along the given edges.

Some candidates still seemed unaware, in part (c)(i), of how to represent the potential increases and decreases **from the initial feasible flow**. It requires forward and backward arrows or a duplicate edge: one showing potential forward flow, the other the potential

backward flow. The initial flow values are best written in black ink close to the arrows so that any adjustments can best be shown in pencil so as not to obliterate the initial flow figures. In part (c)(ii), a table was provided so that a flow of 1 along *SABT*, for example, could be listed in the table. The potential forward and backward flows along *SA*, *AB* and *BT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. If candidates obliterate their values of the potential increases and decreases from part (c)(i), then they risk losing marks for that part. The majority of candidates found only 3 augmenting paths and were not able to find the correct maximum flow.

In part (d), those who used flow augmentation correctly usually had no trouble in completing the diagram to show a maximum flow of 44 litres per second.

In part (e), very few candidates realised the need to consider their saturated arcs in order to obtain a minimum cut. Many wrongly considered their flow on Figure 3, where obviously every cut has a value of 44, and consequently wrote down cuts such as *BT*, *ET* and *GT* as their final answer.

Q12.

Part (a) All candidates seemed familiar with the idea of a super-source, *S*, and super-sink, *T*. Although it was possible to assign equal values such as 100 to the capacity on each of the new edges, such a large value often caused problems with flow augmentation. Some incorrectly assigned equal values such as 20 to the new edges. Those using capacities of 22 on *SP* and 15 on *SQ* were usually the most successful.

Part (b) Quite a few candidates made errors in calculating the value of the cut with a common wrong answer being 36. Several incorrectly thought that the maximum flow was equal to the value of this cut.

Part (c) The values of the initial maximum flows along the given edges were usually correct.

Part (d) It was good to see candidates trying to set out their solution in a logical manner with most candidates showing potential forward and backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values, particularly the initial flow, should be clear to the examiner when such modification is made. It is not wise to totally obliterate previous values when adjustments are made.

Part (e) Some candidates misread this last part and stated the maximum flow through the network. Very few were able to use their final maximum flow diagram to interpret the correct flow from *Y* to *X*.

Q13.

In part (a), a number of candidates made errors in calculating the value of the cut; a common wrong answer was 39. Several incorrectly thought that the maximum flow was equal to the value of this cut. In part (b), the values of the initial maximum flows were usually correct, and in part (c), it was pleasing to see candidates trying to set out their solution in a logical manner. Some candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made and the final backward flows should be the values transferred onto Figure 1 to give a possible maximum flow. Although quite a few were successful in finding

a cut of the same value as the maximum flow, they failed to convince examiners that they had proved that the flow must be a maximum.

Q14.

In part (a), very few candidates seemed prepared for a question with lower capacities as well as upper capacities, and consequently the correct value of the cut was rarely seen.

In contrast, in part (b), almost everyone scored full marks for the missing flows along the given edges.

In part (c)(i), many candidates seemed unaware of how to represent the potential increases and decreases from the initial flow. It requires forward and backward arrows or a duplicate edge: one showing the potential forward flow, the other the potential backward flow. In part (c)(ii), flow augmentation was not widely understood. A table was provided so that a flow of 2 along *SMNT* could be listed in the table. The potential forward and backward flows along *SM*, *MN* and *NT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. Candidates who used flow augmentation correctly usually had no trouble in completing the diagram in part (c)(iii) to show a maximum flow of 24.

Q15.

The insert seemed to help candidates to set out their solution in a logical manner. Most candidates completed the first network indicating the feasible flow. When doing the flow augmentation in part (b)(i), candidates rarely showed their working properly. It is in cases like this, with minimum and maximum capacities, that the formal method (i.e. two arrows and numbers for each edge showing potential increases and decreases of capacity) can be very helpful for candidates and they should be encouraged to always use it. Teachers might again find it valuable to consult the solution in the mark scheme and to encourage their students to take this approach. Also, in trying to find a cut of 14, only a minority of candidates used the original capacities in the network printed in the question paper.

Q16.

Part (a) was usually correctly answered but a number made errors in calculating the value of the cut in part (b). The values of the initial maximum flows in part (c) were usually correct also. The insert **Figure 4** was intended to help candidates to set out their solution to part (d) in a logical manner. Some candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. It should be clear to the examiner what the previous values were when such modification is made and the final backward flows should be the values transferred onto **Figure 5** to give a possible maximum flow. Very few were successful in finding a cut of the same value as the maximum flow. The final part (e) was well answered by most of those who had found the correct maximum flow.

Q17.

Part (a) A very common wrong answer for the value of the cut in part (i) was 44. This arose from assuming that the edge *VY* contributed 6 rather than zero to the value of the cut. A number of candidates wrote in part (ii) that the maximum flow was equal to the value of their cut and this response scored no marks.

Part (b) It was pleasing to see almost all the candidates finding the correct maximum possible flows along the two routes.

Part (c) The labelling procedure involves showing the flow already in the network, usually as a backward flow, together with the surplus potential flow as two distinct figures on the diagram. Candidates would benefit from seeing how the mark scheme indicates this with a suggested layout. A table was provided to indicate any flow augmentation and it is hoped that candidates will use such a table wisely in any future question of this type. When using the labelling procedure, the numbers on the network should be lightly crossed through so both the original and adjusted values can be clearly seen by the examiner. The setting out of the solution in part (i) needs much greater attention if candidates are to score full marks for using the labelling procedure in future examinations.

The value of the maximum flow in part (ii) was usually stated correctly but not always shown correctly on the diagram in the insert. Although a few made no attempt at the minimum cut, most were able to find this correctly. Some indicated this cut on a diagram; others indicated that the cut passed through the edges UV, UY, XV, XY ; still others listed the two sets of vertices {S, U, x} and {V, Y, W, Z, T} separated by the cut; each of these methods scored full marks.

Q18.

Good candidates were able to score the majority of marks on this question, with clearly labelled diagrams. Weaker candidates failed to score the marks in part (a). Candidates should be encouraged to list the paths that they are augmenting at each stage. Part (c) was the least well answered question on the paper.

Q19.

This was the first time a question had been set with multiple sources. Parts (a) and (b) were, in general, well answered. Nearly all candidates were able to find the value of the cut X in part (c). In part (d)(i) candidates only scored full marks if they showed each separate flow augmentation.

Q20.

More able candidates were able to score maximum marks on this question, with clearly labelled diagrams. Weaker candidates failed to score the marks in parts (b) and (c). There was a lack of understanding of 'an initial flow' and in flow augmentation candidates concentrated their efforts on either the sources or the sinks but not both. Candidates should have listed the paths that they were augmenting at each stage. This would have helped them to establish the additional flow of 22 units that was required in this question.

Q21.

Nearly all the candidates were able to find the value of C_1 correctly but there were very few correct answers for C_2 . Part (b) was, in general, well answered. It is helpful for candidates to show each flow augmentation in order to see more easily when an arc has become saturated.

Q22.

Candidates were required to add a super source and a super sink to the given diagram. This was well answered although a number of candidates did not include values on the arcs that at least equalled the total values at points A and B . Part (b) was well answered and to assist candidates they were expected to take these as initial flows in part (c) before using flow augmentation to the diagram. There were many excellent solutions to this question, with working shown on the diagrams, together with a clearly labelled final

solution.

Q23.

Candidates' answers given in part (a) were not very accurate, nor were the descriptions given for no possible flow in part (b). In part (b), candidates merely had to refer to the restrictions at F . The final part of the question was well answered but a small number of candidates simply gave a final flow without any indication of method.

Q24.

All the candidates answered this question well using the diagrams provided as working models. There were many different initial flows but eventually the final solutions became clear.

Q28.

Although candidates generally understood what was required, the presentation of the answers was very muddled and hard to follow. In this sort of question, candidates should clearly indicate their final answer, possibly by ringing the flow in each arc or by drawing a separate figure for the final answer.

Q30.

Most candidates successfully produced the flow and cut of 7, but sometimes it was very hard to follow the working or the final answer in producing the flow. Some candidates followed the good practice of giving their final answer in a diagram separate from the one in which they did their calculations.

Q31.

Most candidates were able to find a maximum flow and minimum cut in parts (a) and (b). They were also able to augment the flow by 50 when an extra arc was added in part (c). However, very surprisingly, few realised that the neatest way of showing that 700 was the new maximum flow was to show that the new network had a cut of 700.

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